Vision of the Department

To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society.

Mission of the Department

To nurture computer engineering professionals by offering: Dynamic learning environment that encourages problem-solving using emerging technology, · National Education Policy aligned evidence-based technical education, fostering innovation, research, and industry–institute collaboration, Holistic, ethical and sustainable development for lifelong success.

Session 2025-2026(EVEN)

Vision: Dream of where you want.	**Mission:** Means to achieve Vision

Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	**Preparation**	**P: Preparation**	**Pep-CL abbreviation pronounce as Pep-si-lL easy to recall**
PEO2	**Core Competence**	**E: Environment (Learning Environment)**	
PEO3	**Breadth**	**P: Professionalism**	
PEO4	**Professionalism**	**C: Core Competence**	
PEO5	**Learning Environment**	**L: Breadth (Learning in diverse areas)**	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:
Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

"I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life." *to contribute to the development of cutting-edge technologies and Research*.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

SHEIKH WASIMUDDIN

Name and Signature of Student and Date
(Signature and Date in Handwritten)

Shiky
6/3/26

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Session	**2024-25 (EVEN)**	**Course Name**	**Mobile and Web Application for IOT Lab**
Semester	**6**	**Course Code**	**23IOT1624**
Roll No	**63**	**Name of Student**	**SHEIKH WASIMUDDIN**

Practical Number	**04**
Course Outcome	1. Understand and apply basic Kotlin programming constructs including variables, control structures, functions, null safety, and object-oriented concepts for Android app development. 2. Design and develop responsive user interfaces using XML layout files, ViewGroups, and widgets in Android Studio, adhering to Material Design principles. 3. Implement event handling and application logic using Kotlin to capture user input, manage app state, navigate between screens, and interact with Android system components like Toasts and Dialogs 4. Integrate learned concepts to develop a mini Android application that demonstrates use of ViewModel, multi-screen navigation, and proper UI/UX design for real-world problem-solving
Aim	Understand Android Studio environment and project structure.
Problem Definition	Create Android project, analyze manifest, Gradle files, SDK versions, permissions
Theory (100 words)	**1. Android Project** An Android project is the development environment used to create mobile applications using Android Studio. It contains all the components required to design, build, and run an application. When a developer creates a new project, Android Studio automatically organizes files into different folders such as source code, resources, and configuration files. The **java or kotlin folder** stores the program logic, while the **res folder** contains UI layouts, images, and other resources. This structured organization helps developers manage the application easily during development. **2. AndroidManifest File** The **AndroidManifest.xml** file is a key configuration file in every Android application. It provides important information about the app to the Android system before the application runs. The file lists all major application components including activities, services, broadcast receivers, and content providers. It also declares the permissions required by the application, such as access to the

internet, camera, or device storage. In addition, the manifest file defines the application icon, name, and the activity that launches when the app starts.

3. Gradle Files
Gradle files are responsible for handling the **build system** of an Android application. They control how the source code is compiled and converted into a runnable application package such as an APK or App Bundle. Gradle also manages external libraries and dependencies that developers include in their projects. Typically, there are two main Gradle files: one at the **project level** and another at the **module level**. These files allow developers to configure plugins, manage versions, and define different build configurations like debug or release.

4. SDK Versions
SDK versions determine which versions of the Android operating system are supported by an application. Developers define three main SDK values in the project configuration. The **minimum SDK** specifies the lowest Android version that can run the app. The **target SDK** indicates the Android version the application is designed and tested for. The **compile SDK** is the Android version used by the developer tools to compile the code. Setting appropriate SDK values ensures that the application works correctly on multiple Android devices.

5. Permissions
Permissions are security rules in Android that control how applications access device features or user data. If an application needs to use functions such as internet connectivity, location services, camera, or storage, it must request permission from the system. These permissions are declared in the **AndroidManifest.xml** file. Some permissions are automatically granted by the system, while others require the user's approval during runtime. This mechanism protects user privacy and ensures that applications only access the resources they genuinely need.

Procedure and Execution (100 Words)	First, launch Android Studio and click on New Project. Select a suitable project template such as Empty Activity. After selecting the template, enter the required project information including the application name, package name, project location, and minimum SDK version. Once the project is created, review the project layout in the Project panel to understand the different folders and files generated by Android Studio. Next, open the AndroidManifest.xml file to observe the basic configuration of the application, including declared components and required permissions. Then examine the Gradle build files to understand how dependencies and build settings are managed in the project. Check the SDK configurations defined in the module-level Gradle file to confirm the compile SDK, target SDK, and minimum SDK versions. If the application requires access to device features such as the internet, necessary permissions can be added inside the AndroidManifest.xml file. After completing these configurations, build the project and execute the application using either an Android emulator or a connected physical device to verify that the application runs successfully. Steps 1. Install and launch Android Studio on the computer. 2. Click Create New Project and select the Empty Activity template. 3. Provide the required details such as application name, package name, and project directory. 4. Choose an appropriate minimum SDK version for device compatibility. 5. Review the project structure which includes folders like java/kotlin, res, and Gradle scripts. 6. Open the AndroidManifest.xml file to examine the application configuration and permissions. 7. Study the Gradle build files to understand dependency handling and build settings. 8. Confirm the compile SDK, target SDK, and minimum SDK values in the module Gradle file.

9. Add any necessary permissions in the manifest file if the application requires them.
10. Finally, build and run the project on an emulator or a real Android device to check proper execution.

Code:

1. Manifest

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
   xmlns:tools="http://schemas.android.com/tools">

   <application
      android:allowBackup="true"
      android:dataExtractionRules="@xml/data_extraction_rules"
      android:fullBackupContent="@xml/backup_rules"
      android:icon="@mipmap/ic_launcher"
      android:label="@string/app_name"
      android:roundIcon="@mipmap/ic_launcher_round"
      android:supportsRtl="true"
      android:theme="@style/Theme.Practical4_iot63">
      <activity
         android:name=".MainActivity"
         android:exported="true">
         <intent-filter>
            <action android:name="android.intent.action.MAIN" />

            <category android:name="android.intent.category.LAUNCHER"
/>
         </intent-filter>
      </activity>
   </application>

</manifest>
```

2. Build.gradel.project

```
// Top-level build file where you can add configuration options common to all sub-
projects/modules.
plugins {
   alias(libs.plugins.android.application) apply false
}
```

3. Build.gradel.app

```
plugins {
   alias(libs.plugins.android.application)
}

android {
   namespace = "com.example.practical4_iot63"
   compileSdk {
      version = release(36) {
         minorApiLevel = 1
      }
   }

   defaultConfig {
      applicationId = "com.example.practical4_iot63"
      minSdk = 24
      targetSdk = 36
      versionCode = 1
      versionName = "1.0"

      testInstrumentationRunner =
"androidx.test.runner.AndroidJUnitRunner"
   }

   buildTypes {
      release {
         isMinifyEnabled = false
         proguardFiles(
            getDefaultProguardFile("proguard-android-optimize.txt"),
            "proguard-rules.pro"
         )
      }
   }
   compileOptions {
      sourceCompatibility = JavaVersion.VERSION_11
      targetCompatibility = JavaVersion.VERSION_11
   }
}

dependencies {
   implementation(libs.androidx.core.ktx)
   implementation(libs.androidx.appcompat)
   implementation(libs.material)
```

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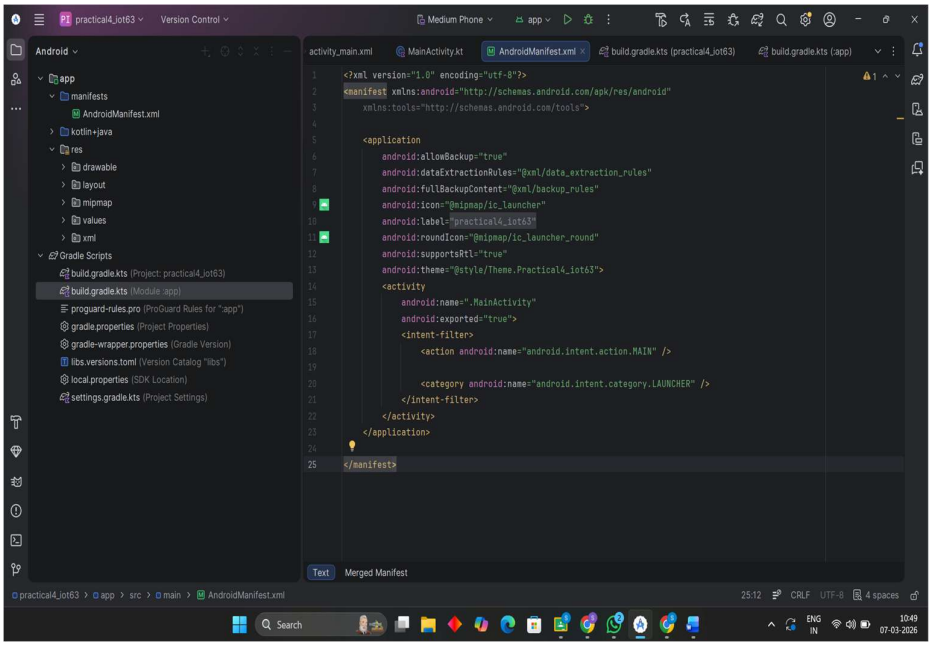
```
    implementation(libs.androidx.activity)
    implementation(libs.androidx.constraintlayout)
    testImplementation(libs.junit)
    androidTestImplementation(libs.androidx.junit)
    androidTestImplementation(libs.androidx.espresso.core)
}
```

Output:



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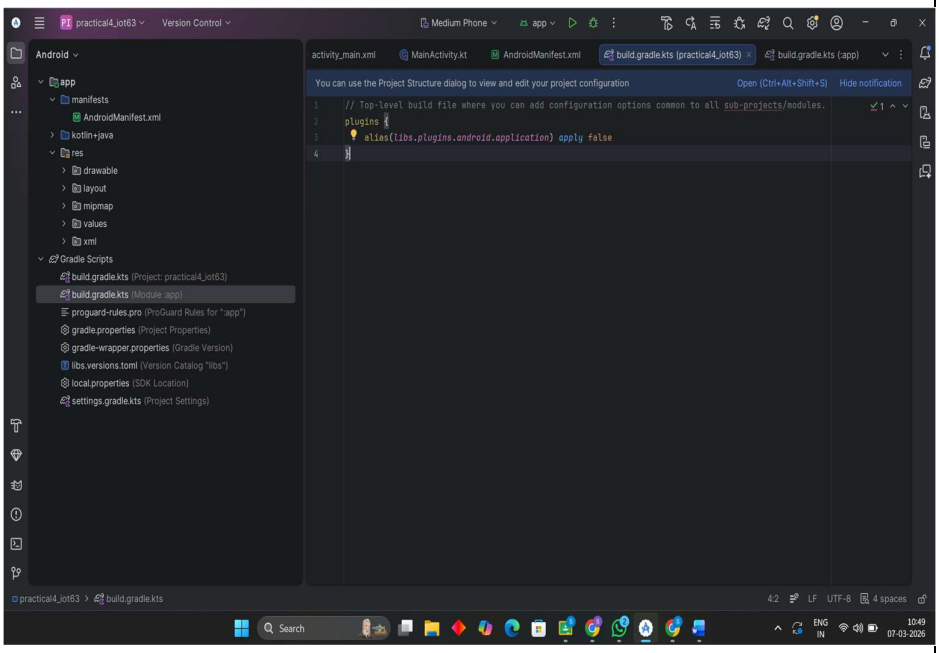

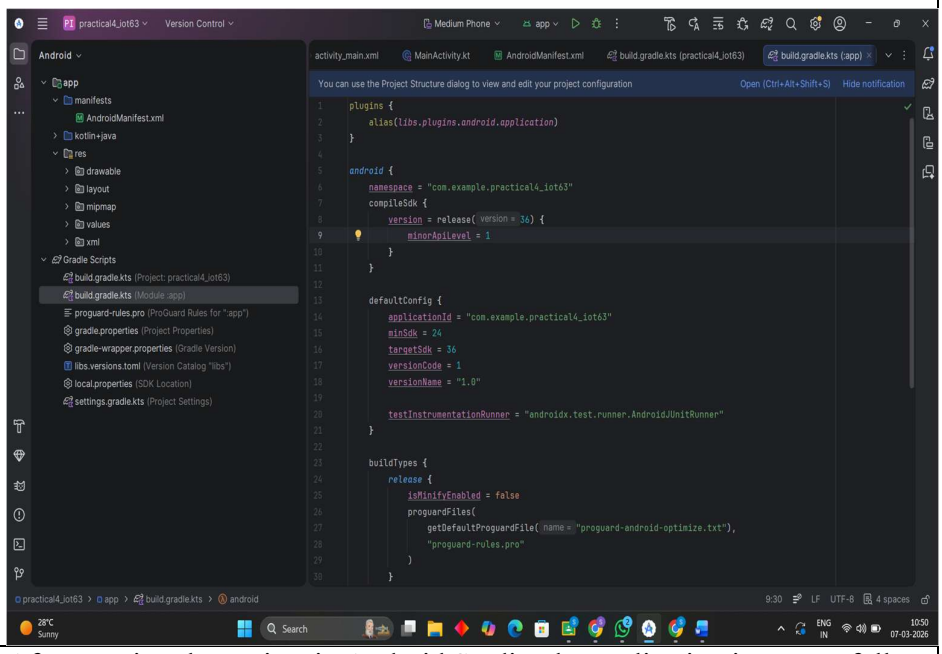

Output Analysis	After running the project in Android Studio, the application is successfully built and executed on the emulator or Android device. The project structure is clearly organized into different folders such as source code, resources, and Gradle scripts. The AndroidManifest.xml file displays the application components and declared permissions. The Gradle files manage dependencies and build configurations effectively. The SDK versions

	ensure compatibility with the selected Android platform. If permissions are added, they are correctly recognized by the system. This process confirms that the Android development environment, project structure, and configuration files are properly set up and functioning as expected.
Link of student Github profile where lab assignment has been uploaded	https://github.com/sheikhwasimuddin
Conclusion	From this experiment, the Android Studio environment and the basic structure of an Android project were successfully understood. The important configuration files such as AndroidManifest.xml, Gradle build files, and SDK settings were analyzed to understand their roles in application development. Permissions required by the application were also studied and configured. This helped in understanding how Android applications are organized and built. Overall, the experiment provided practical knowledge about setting up an Android project and managing its essential components effectively within the Android Studio development environment.

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Date	**06/03/2026**